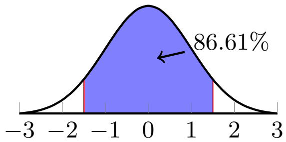


Unless stated otherwise, all problems are worth equal value and answers should be stated as decimals rounded to three decimal places. You must show your work to receive credit.

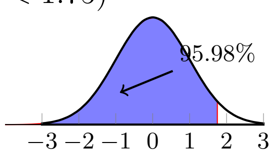
1. A population is normally distributed with mean 24.7 and standard deviation 2.3.

a. Draw a sketch of the bell curve labeling the mean and the intervals representing 1, 2, and 3 standard deviations of the mean.

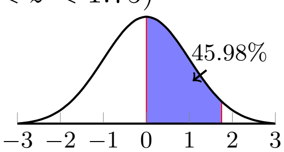
b. Find the percentage of data lies in each of the intervals.

2. Find the following probabilities,

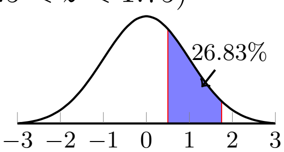
a. $p(z < 1.75)$



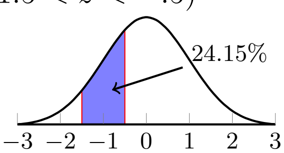
b. $p(0 < z < 1.75)$



c. $p(-.5 < z < 1.75)$

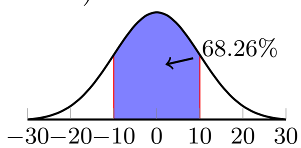


d. $p(-1.5 < z < -.5)$



3. A population is normally distributed with a mean 250 and standard deviation 24. Find the following probabilities.

a. $p(x < 200)$

i. 

b. $p(200 < x < 0)$

c. $p(200 < x < 300)$

d. $p(100 < x < 200)$